Tech Products & Brand Perception

Public Opinion & Sentiment Analysis Project

Juan Diego Grajales Castillo

Carmen Sofía Flórez Juajibioy

Edgar Alejandro Mora Chala

Course: Data Analysis Programming

March 28, 2026

1 Introduction

In today's digital era, consumer perception plays a fundamental role in the success of technological products. Opinions shared on digital platforms such as social media and specialized websites directly influence brand reputation and purchasing decisions.

This project proposes the development of a solution based on data engineering and Natural Language Processing (NLP) to analyze public perception of technology products. By integrating multiple data sources, the goal is to identify opinion patterns, trends, and user satisfaction levels.

2 Justification

The exponential growth of user-generated content has led to large volumes of unstructured data that are difficult to analyze manually. In the technology sector, where competition is high and innovation is constant, understanding user perception is essential for improving products and market strategies.

This project enables the transformation of scattered data into useful information through automated analysis techniques, facilitating data-driven decision-making. Additionally, it integrates different sources to provide a more comprehensive view of consumer behavior.

3 Objectives

3.1 General Objective

To design a data engineering and Natural Language Processing pipeline that enables sentiment analysis of technology products and brands using multiple data sources.

3.2 Specific Objectives

- Collect data from a social media API and through web scraping techniques.
- Process and clean the data for further analysis.
- Apply sentiment analysis techniques to the collected data.
- Design a layered data architecture (Bronze, Silver, Gold).
- Generate visualizations to interpret the obtained results.

4 Theoretical Framework

Sentiment analysis is a Natural Language Processing (NLP) technique used to determine the polarity of a text, classifying it as positive, negative, or neutral. It is widely used to analyze user opinions on social media and digital platforms.

On the other hand, data engineering focuses on designing systems capable of collecting, storing, and processing large volumes of data. Within this context, Data Lake architectures allow information to be organized into different processing levels.

The Bronze, Silver, and Gold layered model is commonly used in data pipelines. The Bronze layer stores raw data, the Silver layer performs cleaning and transformation processes, and the Gold layer contains data ready for analysis and visualization.

5 Methodology

The project is based on the construction of a data pipeline that integrates different sources of information. First, data is collected from the API of the platform X (formerly Twitter) and through web scraping from *GSMArena*.

The collected data is stored in JSON format in the Bronze layer. Then, cleaning, normalization, and transformation processes are applied in the Silver layer, including NLP techniques such as tokenization and stop-word removal.

Finally, in the Gold layer, aggregated metrics and results are generated to enable sentiment analysis and trend visualization.

6 Data Sources

The project uses two main data sources. The first is the API of platform X, which provides real-time user opinions about technology products. The second source is *GSMArena*, a specialized platform for mobile device reviews, from which data is extracted using web scraping techniques.

These sources provide complementary perspectives by combining unstructured social media data with more structured user reviews based on direct product experience.

7 System Architecture

The system architecture is based on a three-layer model:

- **Bronze:** Raw data storage in JSON format.
- **Silver:** Data processing and cleaning.
- **Gold:** Generation of metrics and data ready for visualization.

The system uses Python as the main programming language, along with libraries such as Pandas and BeautifulSoup. Pipeline orchestration is handled using Apache Airflow, and visualization is implemented with Plotly Dash.

8 Project Progress

So far, significant progress has been made in the development of the project. Data sources have been identified and validated, achieving data extraction both from APIs and web scraping.

Additionally, the pipeline architecture and technologies to be used have been defined. Sample datasets have also been collected, allowing validation of the feasibility of sentiment analysis.

9 Preliminary Results

The collected data shows the presence of diverse opinions, including positive, negative, and neutral comments about different technological products.

Challenges related to data quality have been identified, such as multilingual content, informal text, and irrelevant information. This highlights the importance of proper data cleaning and preprocessing techniques.

10 Conclusions

The project demonstrates the feasibility of building a sentiment analysis system based on multiple data sources. The integration of social media data and review platforms provides a more comprehensive understanding of user perception.

Additionally, important challenges in processing unstructured data are identified, which can be addressed through appropriate NLP techniques.

11 Future Work

Future work includes fully implementing the pipeline in Apache Airflow, improving sentiment analysis models, and developing an interactive dashboard for result visualization.